



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2014

BIOLOGY

8875/01

HIGHER 1

19 September 2014

Paper 1 Multiple Choice

1 hour

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, centre number and index number on the Answer Sheet provided.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

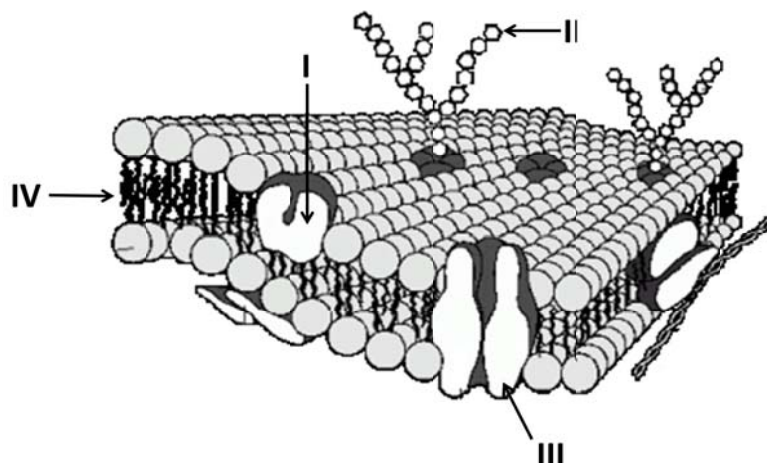
Calculators may be used.

This question paper consists of **17** printed pages.

- 1 A particular genetic disorder causes the progressive degeneration of nerve cells, eventually leading to death. It is caused by a defective enzyme associated with one of the cell's organelles. The enzyme is normally responsible for the breakdown of glycolipids in the cell surface membrane. In the absence of the functional enzyme, accumulation of lipids within the nerve cells leads to the death of these cells.

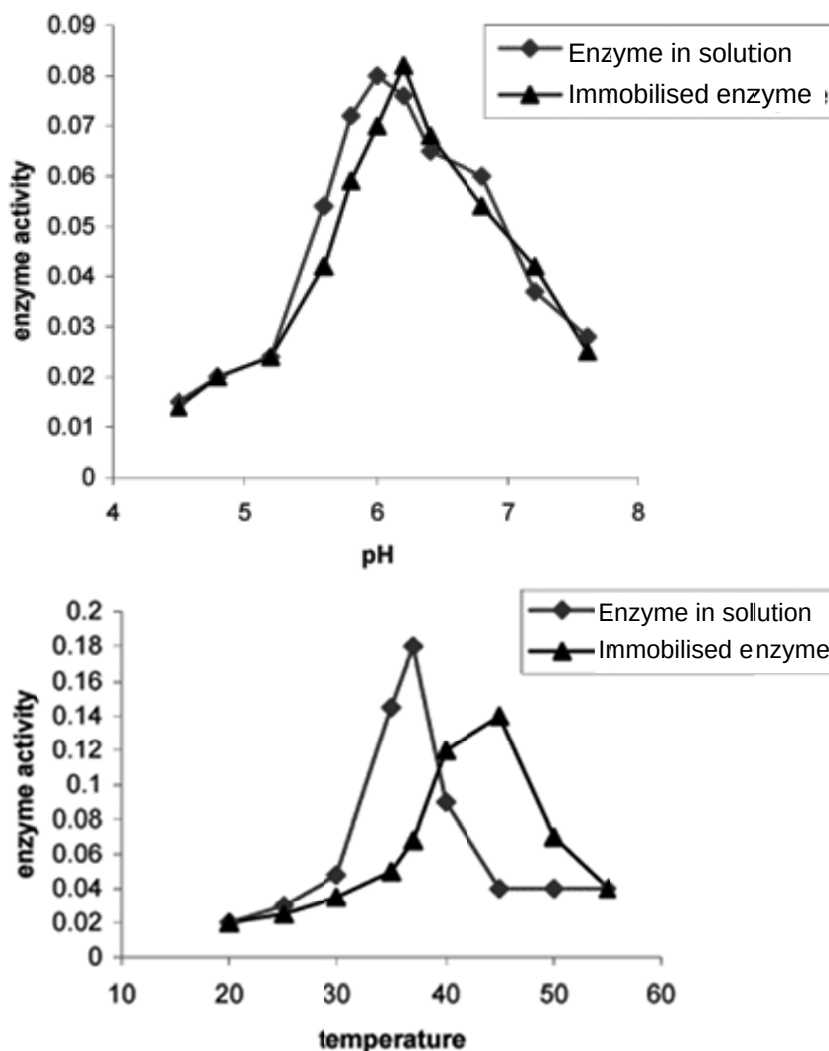
Which organelle is the defective enzyme likely to be associated with?

- A Nucleus
 B Lysosome
 C Rough endoplasmic reticulum
 D Smooth endoplasmic reticulum
- 2 The diagram below shows a section of the cell surface membrane from an arctic fish. Which of the following options regarding the labelled components of the membrane is correct?



	Labelled Cell Component			
	I	II	III	IV
A	Plays a role in cell-cell adhesion	Plays a role in cell-cell recognition	Allows transport of proteins like insulin	Contains a high amount of saturated lipids
B	Plays an enzymatic role	Variation in branching of oligosaccharide allows for cell-cell recognition	Allows transport of amino acids	Contains a high amount of cholesterol
C	Maintains the fluidity of the cell surface membrane	Variation in branching of amino acids allows cell-cell recognition	Plays a role in maintaining membrane potential	Is produced at the Rough Endoplasmic Reticulum
D	Is involved in active transport of ions	Is termed collectively with other glycolipids as the glycocalyx	Is localised to a specific region in the cell surface membrane	Plays a role in cell-cell recognition

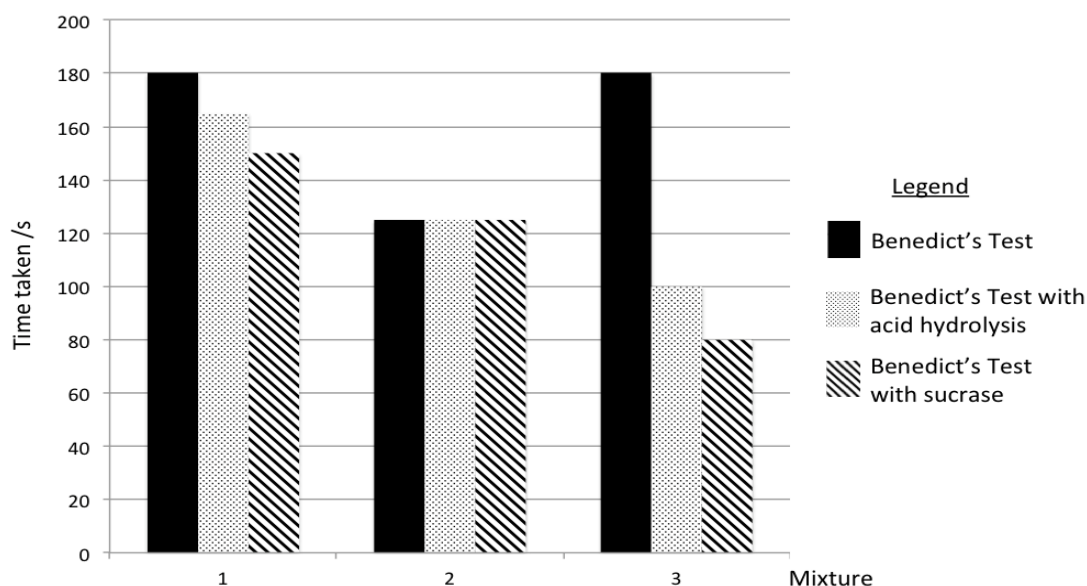
- 3 When used industrially, enzymes are often immobilised on a solid support so that they can be easily removed from the reaction mixture and reused. The following graphs compare the activity of an enzyme in solution against the same enzyme immobilised on a solid support, under different pH and temperature.



Which of the following statements is true?

- A Enzyme activity is about the same at low pH for both the immobilised enzyme and the enzyme in solution but the optimum pH is significantly higher for the immobilised enzyme as it is more resistant to pH changes.
- B At 20°C, the 3D configuration of both the immobilised enzyme and the enzyme in solution is altered, thus accounting for the low enzyme activity.
- C The bonds in the immobilised enzyme remain intact at a higher temperature than the enzyme in solution, thus accounting for the higher enzyme activity above 40°C.
- D The immobilised enzyme gives a higher $V_{\max}$ value under both optimal pH and temperature than the enzyme in solution.

- 4 Three mixtures (1, 2 and 3) containing different concentrations of glucose and sucrose were given to a student. In order to determine the relative concentration of each substance in the mixture, the student subjected the mixtures to different conditions before carrying out Benedict's Test. The time taken for brick-red precipitate to form was recorded in the graphs below.



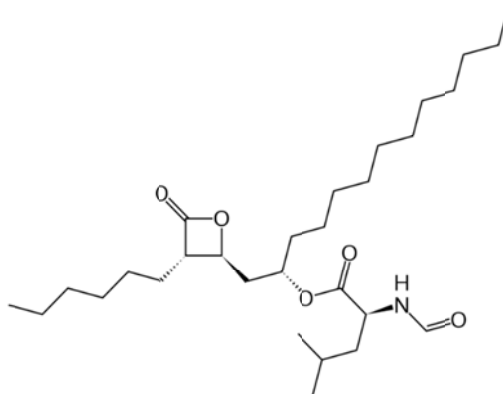
Which of the following statement can be concluded from these graphs?

- A Mixture 2 contains equal concentrations of glucose and sucrose.
 B Mixture 1 has a higher concentration of glucose than Mixture 2.
 C Mixture 3 has a higher concentration of sucrose than Mixture 1.
 D Mixture 3 has a lower concentration of sucrose than Mixture 1.
- 5 To identify an unknown substance **X**, a scientist conducted the following series of tests.

Mixture tested	Test	Result
X mixed with glycogen	Iodine	Red-brown colour
X mixed with starch	Benedict's solution	Orange precipitate
X only	Biuret solution	Violet colour
X only	Ethanol emulsion	Clear

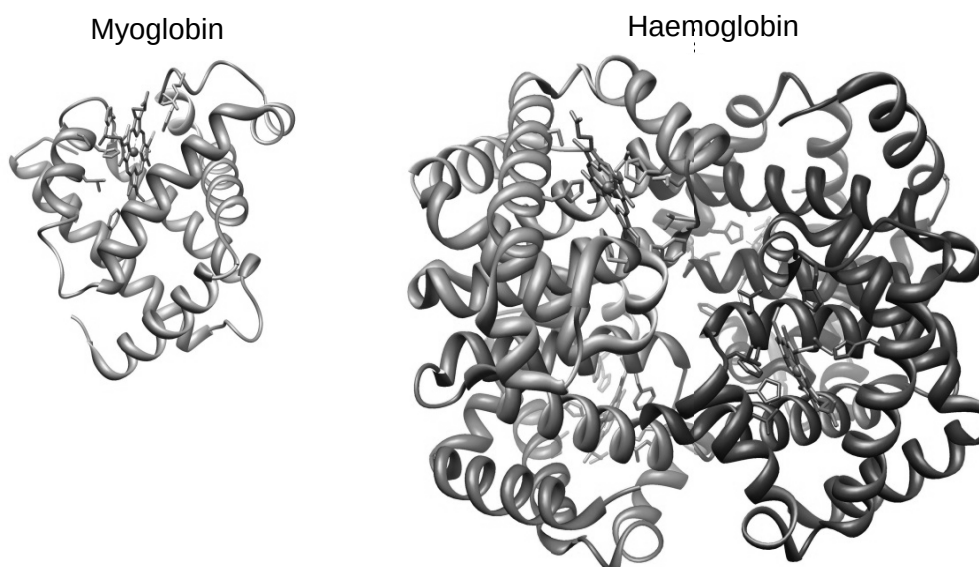
What is substance **X** likely to be?

- A Maltose
 B Amylose
 C Amylase
 D Fatty acid



A Lipase
B Protease
C Carbohydrase
D RNase

7 Myoglobin is a protein that serves a similar function to haemoglobin, but is only found in muscle cells. The structures of both proteins are shown below.



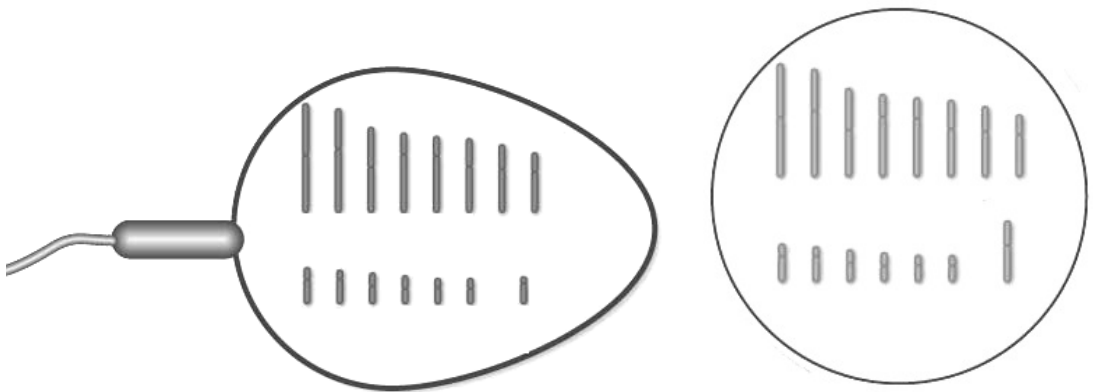
A Four oxygen molecules can bind to myoglobin at the same time.

B Hydrophobic amino acid residues in myoglobin are projected towards the exterior.

C Binding of oxygen to the haem group is reversible.

D Myoglobin exhibits cooperativity in oxygen binding.

- 8 The following statements each describe a similarity or difference between mitosis and meiosis. Which statement is false?
- A Mitosis always produces diploid cells while meiosis always produces haploid cells.
 - B Mitosis always results in two daughter cells while meiosis always results in four daughter cells.
 - C Both mitosis and meiosis can occur in the absence of centrioles.
 - D Both mitosis and meiosis involve the separation of sister chromatids.
- 9 The diagram below shows the egg cell and sperm cell formed by one mammal species, as well as the number of chromosomes they contain.



How many DNA molecules would be found in a somatic cell from this organism at anaphase of mitosis?

- A 15
 - B 30
 - C 60
 - D 120
- 10 The retinoblastoma protein (Rb protein) is coded for by the *RB1* gene. Rb protein prevents a cell from progressing into the S phase of a cell cycle when damaged DNA is detected. When both copies of the *RB1* gene are mutated and dysfunctional, cells with damaged DNA may continue to divide uncontrollably to form a tumour.
- Which of the following statements is true?
- A *RB1* gene is a proto-oncogene.
 - B The mutated *RB1* allele acts in a recessive manner.
 - C The mutated *RB1* allele codes for a hyperactive Rb protein.
 - D A gain-of-function mutation has occurred.

11 The mechanism of action of four drugs that inhibit DNA replication is stated below.

- 1 Aphidicholine inhibits DNA polymerase.
- 2 Cytarabine is converted into a molecule that can substitute for a DNA nucleotide and also inhibits DNA repair mechanisms.
- 3 Epirubicin inhibits an enzyme involved in the unwinding and separation of DNA strands.
- 4 Hydroxycarbamide inhibits an enzyme involved in the production of deoxyribonucleotides.

Which row correctly matches the effects of these drugs on DNA replication?

	Effects of Drug on DNA Replication			
	Decreased pool of available nucleotides inhibits chain elongation	DNA damaged during replication and cell death occurs	DNA strands not available as templates for replication	Exposed DNA template strands unable to be copied
A	aphidicholine	cytarabine	epirubicin	hydroxycarbamide
B	hydroxycarbamide	epirubicin	aphidicholine	cytarabine
C	epirubicin	hydroxycarbamide	cytarabine	aphidicholine
D	hydroxycarbamide	cytarabine	epirubicin	aphidicholine

12 An insertion mutation occurs in the gene coding for an enzyme, tyrosinase. Nucleotide sequences of the gene (the non-template strand), as well as the corresponding amino acid sequence of tyrosinase, are shown below.

Wild-type allele	ATG	AAG	TTG	GCT	AAA	TGG	GGA
Wild-type protein	Met	Lys	Leu	Ala	Lys	Trp	Gly
Mutant allele	ATG	AAG	TTA	GGC	TAA	ATG	GGG
Mutant protein	Met	Lys	Leu	Gly	-		



Insertion of adenine

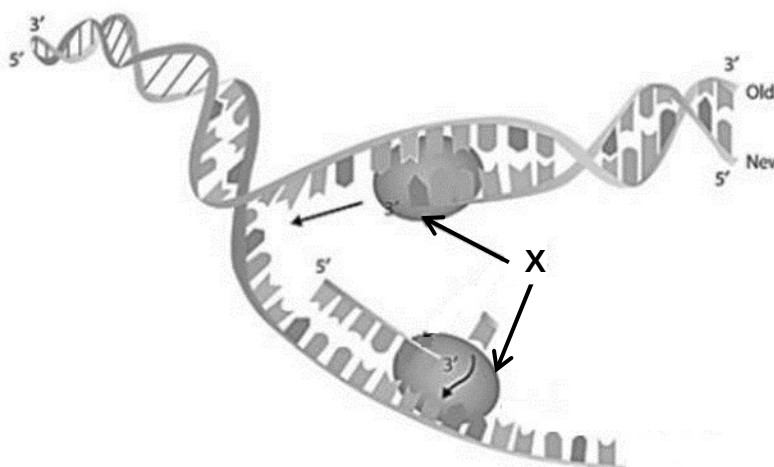
Which feature of the genetic code **cannot** be observed based on the information given?

- A** The genetic code is degenerate.
- B** The genetic code is punctuated.
- C** The code is non-overlapping.
- D** The code is universal.

13 Gene mutations involve changes in the nucleotide sequence of DNA. Which of the following descriptions regarding gene mutations is correct?

- A** Frame-shift mutations can be caused by base substitution.
- B** Missense mutations can be selectively neutral.
- C** Silent mutations may not be selectively neutral.
- D** Frame-shift mutations can result from an inversion of bases.

14 The diagram below represents a process that occurs in prokaryotes.



Which of the following correctly describes the characteristic of **X** for its function?

- A** It reads the template 5' to 3'.
- B** It requires the involvement of other enzymes prior to its action.
- C** Only 1 molecule of **X** can act on each template strand at any one time.
- D** It produces various types of nucleic acids in the process.

15 A couple has three children, all of whom have brown eyes and blond hair. Both parents are heterozygous for brown eyes but one is a blond (rr) and the other is a redhead (Rr).

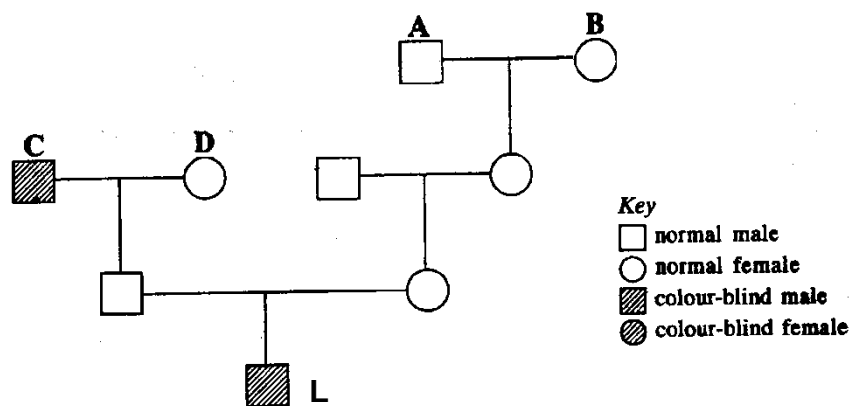
What is the probability that their next daughter will be a brown-eyed redhead?

- A** 1/2
- B** 3/8
- C** 1/16
- D** 3/16

16 Which of the following statements is true?

- A Continuous variation shows a normal distribution and is only influenced by genetic factors.
- B Continuous variation is controlled by many genes and the traits are usually well defined with no gradation.
- C Discontinuous variation shows traits that follow discrete distribution and is mostly influenced by environmental factors.
- D Discontinuous variation shows traits that are controlled by one or two genes and is relatively unaffected by environmental factors.

17 The diagram shows the pedigree of a family carrying the sex-linked allele for colour-blindness.

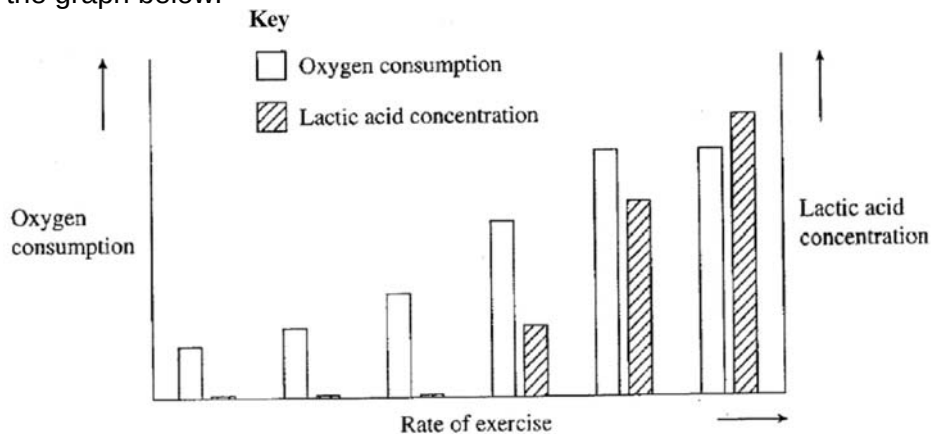


From which labelled member of his family A, B, C or D, did L inherit this disorder?

18 Which of the following statement(s) is/are true of polyploidy?

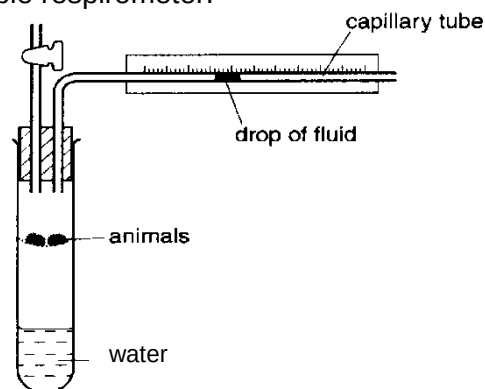
- 1 Condition of the nucleus where there are three or more times the haploid number of chromosomes.
 - 2 It can arise due to non-disjunction of sister chromatids of one or several chromosomes during anaphase within a diploid nucleus.
 - 3 It can arise due to fusion of two haploid gametes.
 - 4 It is much more common in animals than in plants.
- A 1 only
 - B 2 and 3
 - C 1, 2 and 3
 - D 1, 2, 3 and 4

- 19 Scientists measured the oxygen consumption of an athlete at different rates of exercise. They also measured the lactic acid concentration in the athlete's muscles. The results are shown in the graph below.



Which of the following can be deduced from the information in the graph?

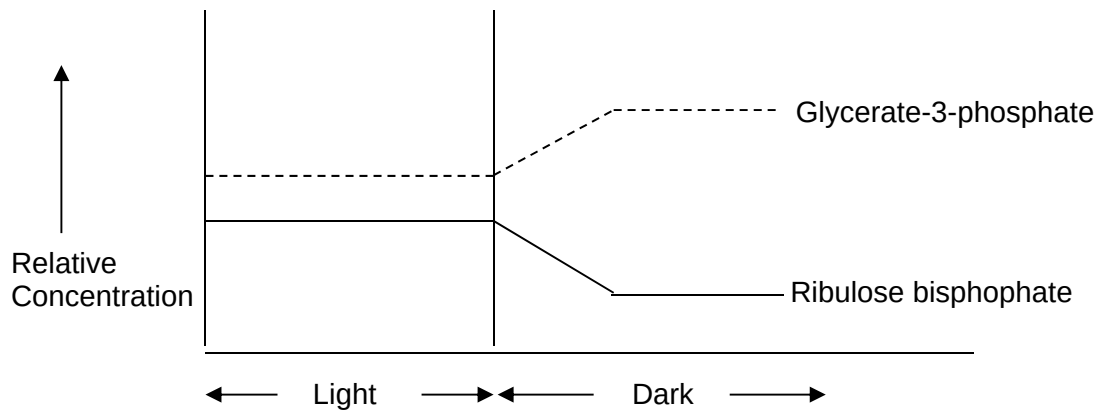
- A Anaerobic respiration occurs at a constant rate.
 - B At a higher rate of exercise, rate of anaerobic respiration in muscle cells increases.
 - C Oxygen consumption increases proportionally with increased rate of exercise.
 - D Increase in lactic acid concentration is due to increased oxygen consumption.
- 20 The diagram shows a simple respirometer.



The drop of fluid in the capillary tube moved to the right after one minute. What can be concluded from the information provided?

- A Anaerobic respiration is occurring.
- B The animals could be using carbohydrates as the respiratory substrate.
- C The animals could be using fats or proteins as the respiratory substrate.
- D The respiratory quotient is 0.9.

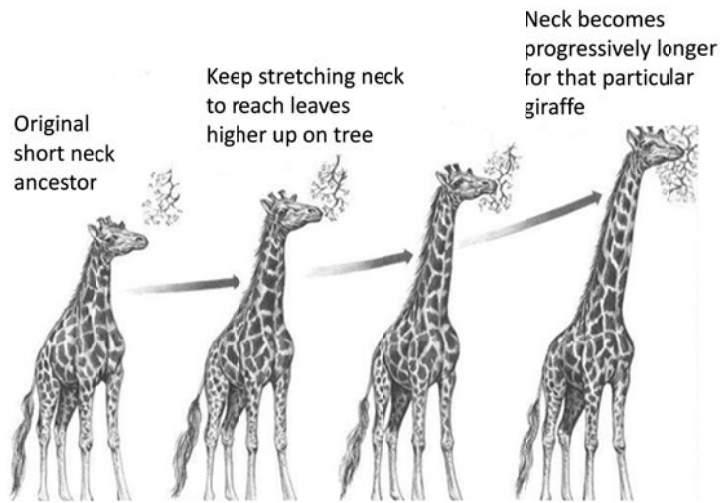
- 21 The graph below shows the effects of a change from light to dark conditions on the relative concentrations of glycerate-3-phosphate (GP) and ribulose biphosphate (RuBP) in a plant?



Which of the following features of the graph does not support the hypothesis that RuBP is converted to GP? The compounds

- A have constant concentrations at the start of the experiment.
 - B respond immediately to the changed condition.
 - C achieve a new steady concentration at the same time.
 - D have the same rate of response, one negative and the other positive.
- 22 Which of the following processes could still occur in a chloroplast in the presence of an inhibitor that prevents H^+ from passing through ATP synthase complexes?
- 1 Sugar synthesis
 - 2 Photolysis of water
 - 3 Transfer of electrons down the electron transport chain
 - 4 Oxidation of reduced NADP
- A 1 and 2
 - B 2 and 3
 - C 3 and 4
 - D 1 and 4

23 The diagram below shows a theory that proposes the evolution of the long neck giraffe.

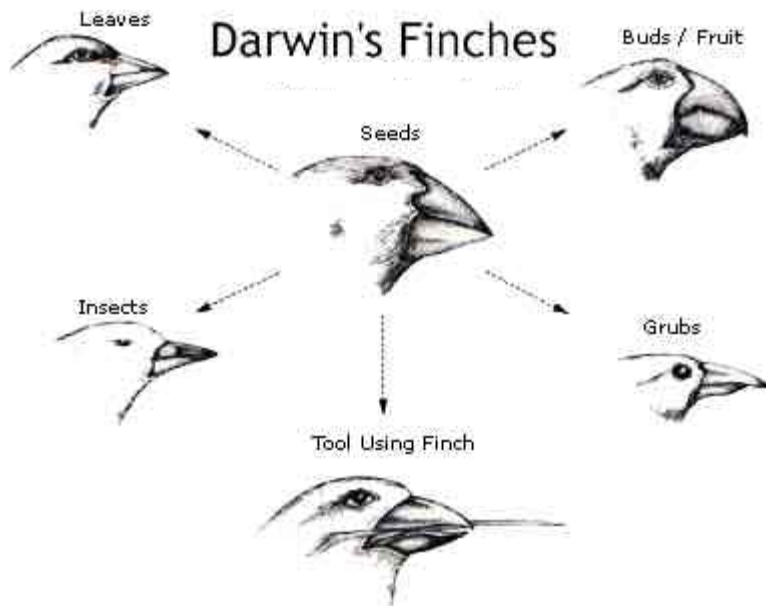


Which of the following statements are correct critiques on the theory based on Neo-Darwinism?

- 1 The length of the giraffe's neck is fixed because it is determined by the environment.
- 2 Variations in the length of the neck due to environmental factors cannot be passed down to its offspring as it is not a heritable change.
- 3 Evolution cannot occur at the individual level.
- 4 The evolution of the long neck is driven by the needs of the giraffe.

- A 1 and 2
 B 2 and 3
 C 1 and 4
 D 2, 3 and 4

- 24 The diagram shows the evolution of different species of Darwin's finches on Galapagos Island from the seed eating ancestor finch.

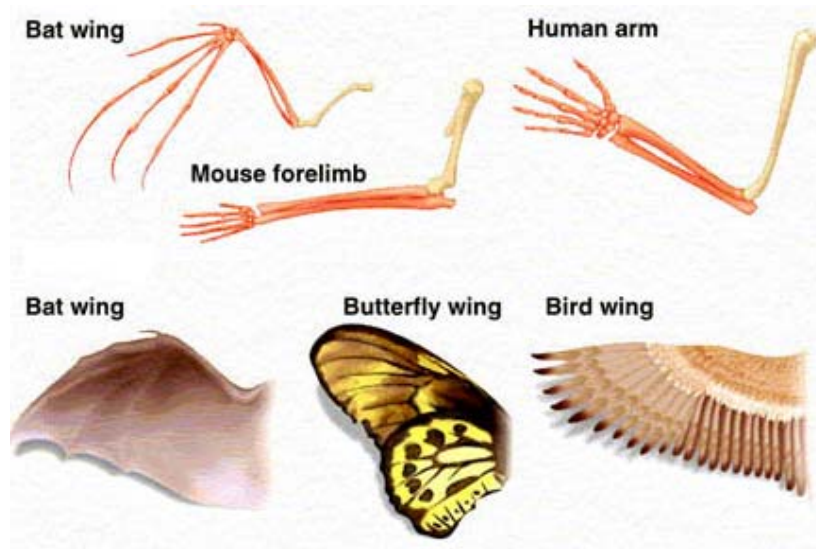


Which of the following statements are correct explanations for the evolution of the finches with different beak size and shapes?

- 1 The ancestor finch develops different beak size and shapes in response to the different environmental pressures.
- 2 Variations in beak size and shapes are present in the ancestor population.
- 3 Different environmental pressure exerts different selection pressure on the ancestor finch at various locations.
- 4 Finches with better adapted beak size and shape to their local environment have higher fitness than others.

- A 1 and 2
 B 2 and 3
 C 1 and 4
 D 2, 3 and 4

- 25 The diagram below shows the bone structures of the human arm, mouse forelimb and bat wings as well as the morphology of the wings of bat, butterfly and birds.



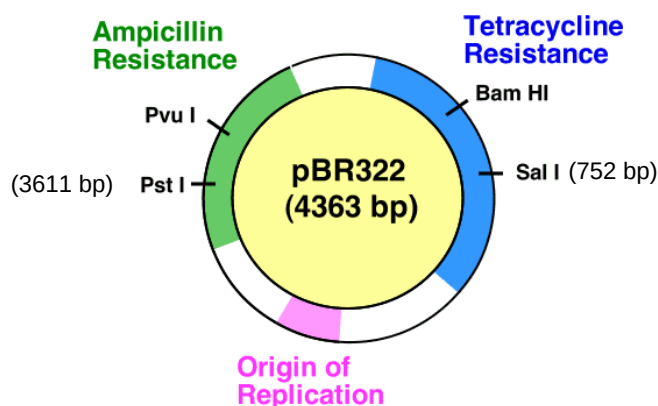
Which of the following are correct conclusions made from the diagram provided?

- 1 Bat, mouse and human share a common ancestor as their bone structure exhibit anatomical homology.
- 2 Variations in the bone morphology of bats, mouse and human are due to natural selection.
- 3 Bat, butterfly and bird share a recent common ancestor as shown by their common wing morphology.
- 4 Bat, butterfly and bird exhibit analogous structures.

- A** 1 and 2
B 2 and 3
C 1 and 4
D 1, 2 and 4

For questions **26** and **27**, please refer to the diagram below which shows an ampicillin-resistant, tetracycline-resistant plasmid, pBR322.

- 26** The figure below shows an ampicillin-resistant, tetracycline-resistant plasmid, pBR322 (not drawn to scale). In the preparation of a *Drosophila* genomic library, pBR322 is cleaved with *Pst*I before it is ligated with *Pst*I-digested *Drosophila* DNA. The resulting mixture is then used to transform *E. coli*.

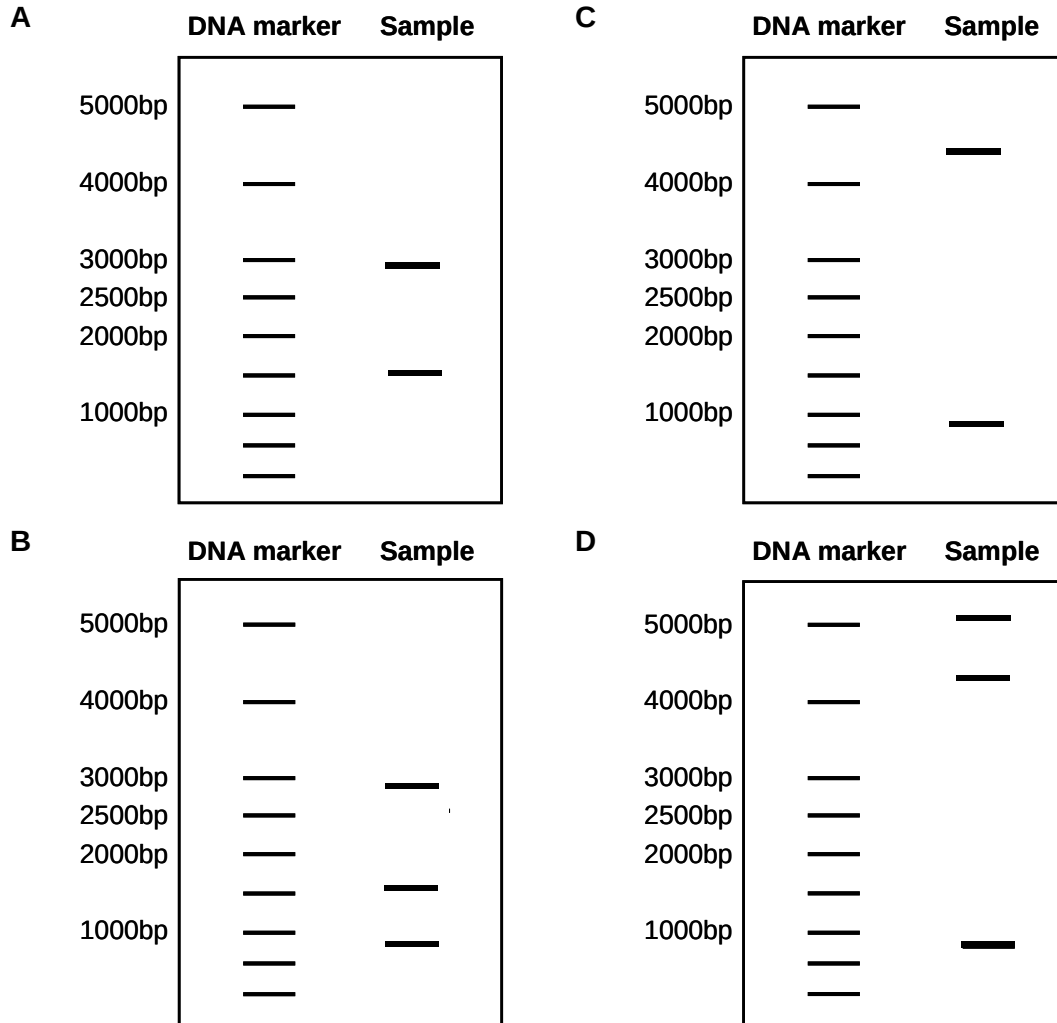


Which antibiotic should be added to the medium to screen for recombinant clones and what growth pattern should be selected to obtain recombinant clones containing *Drosophila* DNA inserts?

	Antibiotic to be added	Growth pattern for selection
A	Ampicillin	Colonies that grow on ampicillin medium should be tested for growth on a tetracycline medium by replica plating. Colonies that failed to grow with tetracycline probably contain the <i>Drosophila</i> DNA insert.
B	Ampicillin	Colonies that failed to grow on ampicillin medium should be transferred to a medium with X-gal by replica plating. White colonies formed should probably contain the <i>Drosophila</i> DNA insert.
C	Tetracycline	Colonies that grow on tetracycline medium should be tested for growth on an ampicillin medium by replica plating. Colonies that failed to grow on ampicillin probably contain the <i>Drosophila</i> DNA insert.
D	Tetracycline	Colonies that failed to grow on tetracycline medium should be transferred to a medium with X-gal by replica plating. White colonies formed should probably contain the <i>Drosophila</i> DNA insert.

- 27 A recombinant plasmid, pBR322, containing the inserted *Drosophila* gene of 726 bp in size, was isolated. It was then subjected to digestion with *Sall* and *PstI*. The DNA fragments are then separated by size through gel electrophoresis and the sizes of the separated fragments are determined by comparison to a DNA marker.

Which of the following gel electrophoregrams shows the correct sizes of the separated fragments?



- 28 Which of the following is **not** a benefit of the Human Genome Project?

- A Identification of genes linked with specific genetic disorders.
- B Screening populations for heterozygous carriers.
- C Improving the understanding of evolutionary links.
- D Producing new drugs that target cancer cells in humans.

29 Which row shows the normal functions of different types of stem cells in humans?

	Functions		
A	Cells that can differentiate into bone cells in the skeleton.	Cells that can differentiate into all types of cells during the first 3 days of embryo development.	Cells that can be used for gene therapy to treat ADA-SCID.
B	Cells that can differentiate into cardiac muscle cells in the heart.	Cells that can differentiate into all types of cells during the first 3 days of embryo development.	Cells that can produce more red blood cells to replace worn out cells.
C	Cells that can produce more insulin in damaged pancreas.	Cells that can differentiate into all types of cells in the blastocyst stage of embryo development.	Cells that can differentiate into cartilage cells in the joints.
D	Cells that can differentiate into nerve cells in the brain.	Cells that can differentiate into all types of cells in the blastocyst stage of embryo development.	Haematopoietic stem cells that can produce differentiated cells in the brain.

30 One type of genetically-modified corn has

- a gene for the production of BT toxin which protects the plant against specific insect;
- a '*pat*' gene for tolerance to the herbicide 'Basta' which is used for selection of transformed plants.
- an '*amp*' gene which was introduced in the plant together with the BT toxin gene.

Which of the following is **not** a valid concern regarding mass propagation of such genetically-modified corn?

- A** The consumption of the toxin produced by the genetically-modified corn may have negative physiological effects in human.
- B** The consumption of new proteins produced by the genetically-modified corn may cause allergic reactions in human.
- C** The '*amp*' gene may be transferred to the cells lining the intestine tract making treatment with ampicillin ineffective for disease caused by bacteria.
- D** Gene transfer by cross breeding with wild plants may result in the emergence of super weeds.

End of Paper